

STD 1: Measurement (Std 1) M4 Rates (Y12)

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Exam Equivalent Time: 108 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

IMPORTANT FEATURES AND TIPS FROM 2UG EXAM HISTORY

- *MS-M4 Rates* has attracted multiple questions in both the 2020 and 2019 Std1 exams.
- *MS-M4 Rates* covers a wide range of content such as best buys, energy, heart rates, medication, speed calculations, and fuel consumption.
- This topic has a significant overlap with *M1 Mass and Energy*, *A1 Algebraic Applications* and *A2 Linear Relationships*. Questions that fit into two categories are included in both by database design.

ANALYSIS - What to Expect and Common pitfalls

- *Rates* is a very broad topic area, although students who have a good conceptual understanding should expect to score highly.
- Speed, distance and time questions (see also *A1 Algebraic Applications*) are well covered in the database, along with fuel consumption (see also *A2 Linear Relationships*) and other related "rate" contexts.
- Distance-time graphs are explicitly mentioned in the Std1 syllabus and were tested in *2019 Std1 18*, producing a mean mark of 42%. Attention warranted here.

Questions

1. Algebra, STD2 A2 2017 HSC 14 MC

Kate is comparing two different models of car. Car *A* uses fuel at the rate of 9 L/100 km. Car *B* uses 3.5 L/100 km.

Suppose Kate plans on driving 8000 km in the next year.

How much less fuel will she use driving car *B* instead of car *A*?

- A.** 280 L
 - B.** 440 L
 - C.** 720 L
 - D.** 1000 L
-

2. Measurement, STD2 M7 2016 HSC 11 MC

The concentration of a drug in a certain medication is 100 mg / 5 mL. A patient is prescribed 2000 mg of the drug.

How much medication should be given to the patient?

- (A)** 4 mL
 - (B)** 25 mL
 - (C)** 100 mL
 - (D)** 400 mL
-

3. Measurement, STD2 M7 2016 HSC 9 MC

An old washing machine uses 130 L of water per load. A new washing machine uses 50 L per load.

How much water is saved each year if two loads of washing are done each week using the new machine?

- (A)** 2600 L
 - (B)** 4160 L
 - (C)** 5200 L
 - (D)** 8320 L
-

4. Algebra, STD2 A2 2018 HSC 5 MC

The driving distance from Alex's home to his work is 20 km. He drives to and from work five times each week. His car uses fuel at the rate of 8 L/100 km.

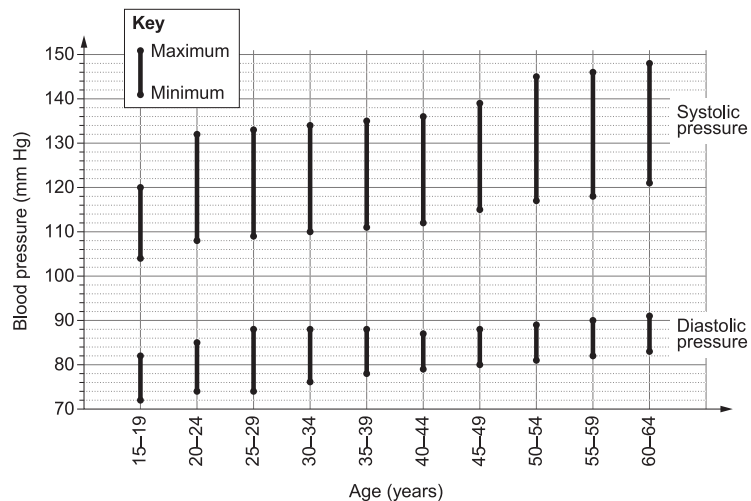
How much fuel does he use driving to and from work each week?

- A. 16 L
- B. 20 L
- C. 25 L
- D. 40 L

5. Measurement, STD1 M4 2019 HSC 6 MC

When blood pressure is measured, two numbers are recorded: systolic pressure and diastolic pressure. If the measurements recorded are 130 systolic and 85 diastolic, then the blood pressure is written as '130 over 85'.

The bars on the graph indicate the healthy ranges of blood pressure for people of various ages.



Which person has both blood pressure measurements in the healthy range for their age?

- A. Stella aged 23 with blood pressure 120 over 72
- B. Shane aged 35 with blood pressure 124 over 90
- C. Jon aged 54 with blood pressure 137 over 94
- D. Annie aged 61 with blood pressure 142 over 88

6. Measurement, STD2 M4 2019 HSC 3 MC

Sugar is sold in four different sized packets.

Which is the best buy?

- A. 100 g for \$0.40
- B. 500 g for \$1.65
- C. 1 kg for \$3.50
- D. 2 kg for \$6.90

7. Algebra, STD2 A2 SM-Bank 4 MC

A car travels 350 km on 40 L of petrol.

What is its fuel consumption?

- A. 7.8 L/100 km
- B. 8.4 L/100 km
- C. 8.8 L/100 km
- D. 11.4 L/100 km

8. Algebra, STD2 A2 2014 HSC 22 MC

Heather's car uses fuel at the rate of 6.6 L per 100 km for long-distance driving and 8.9 L per 100 km for short-distance driving.

She used the car to make a journey of 560 km, which included 65 km of short-distance driving.

Approximately how much fuel did Heather's car use on the journey?

- (A) 37 L
- (B) 38 L
- (C) 48 L
- (D) 50 L

9. Measurement, STD2 M7 2014 HSC 17 MC

A child who weighs 14 kg needs to be given 15 mg of paracetamol for every 2 kg of body weight.

Every 10 mL of a particular medicine contains 120 mg of paracetamol.

What is the correct dosage of this medicine for the child?

- (A) 5.6 mL
- (B) 8.75 mL
- (C) 11.43 mL
- (D) 17.5 mL

10. Measurement, STD2 M7 SM-Bank 15 MC

A brand of pasta is sold in two different packets.



What is the difference in the price **per kilogram** of these two packets?

- A. \$1.50
- B. \$3.00
- C. \$3.60
- D. \$7.20

11. Measurement, STD2 M7 2016 HSC 15 MC

Calls on a mobile phone plan are charged at the rate of 54 cents per 30 seconds, or part thereof.

What is the cost of a call lasting 2 minutes and 15 seconds?

- (A) \$2.16
- (B) \$2.32
- (C) \$2.43
- (D) \$2.70

12. Measurement, STD2 M1 2014 HSC 20 MC

In a household of 4, each member uses an average of 13 minutes of hot water per day.

The household uses a 9 kW hot water unit.

Electricity is charged at 11.97 c/kWh when the hot water unit is being used.

What is the electricity cost for the hot water used by this household in one week?

- (A) \$1.63
- (B) \$6.54
- (C) \$392.14
- (D) \$653.56

13. Measurement, STD1 M4 2019 HSC 5 MC

Which expression can be used to convert a speed of 3 metres per minute to a speed in centimetres per second?

- A. $3 \times 100 \div 60$
- B. $3 \times 100 \times 60$
- C. $3 \div 100 \div 60$
- D. $3 \div 100 \times 60$

14. Measurement, STD2 M7 2018 HSC 27a

Jenny used her mobile phone while she was overseas for one month.

Her mobile phone plan has a base monthly cost of \$50. While overseas, she is also charged 33 cents per SMS message sent and 26 cents per MB of data used.

During her month overseas, Jenny sent 120 SMS messages and used 1400 MB of data.

What was her mobile phone bill for the month overseas? (2 marks)

15. Algebra, STD2 A2 2016 HSC 26c

Peta's car uses fuel at the rate of 5.9 L /100 km for country driving and 7.3 L /100 km for city driving. On a trip, she drives 170 km in the country and 25 km in the city.

Calculate the amount of fuel she used on this trip. (2 marks)

16. Algebra, STD2 A2 2014 HSC 27b

Xuso is comparing the costs of two different ways of travelling to university.

Xuso's motorcycle uses one litre of fuel for every 17 km travelled. The cost of fuel is \$1.67/L and the distance from her home to the university car park is 34 km. The cost of travelling by bus is \$36.40 for 10 single trips.

Which way of travelling is cheaper and by how much? Support your answer with calculations. (2 marks)

17. Measurement, STD2 M7 2015 HSC 27b

A patient requires 2400 mL of fluid to be delivered at a constant rate by means of a drip over 12 hours. Each mL of fluid is equivalent to 15 drops.

How many drops per minute need to be delivered? (2 marks)

18. Measurement, STD2 M1 2016 HSC 28b

The cost of buying a new heater is \$990. It has an energy consumption of 505 kWh per year.

Energy is charged at the rate of \$0.35 kWh.

How much will it cost in total to purchase and then run this heater for five years? (2 marks)

19. Measurement, STD2 M7 SM-Bank 8

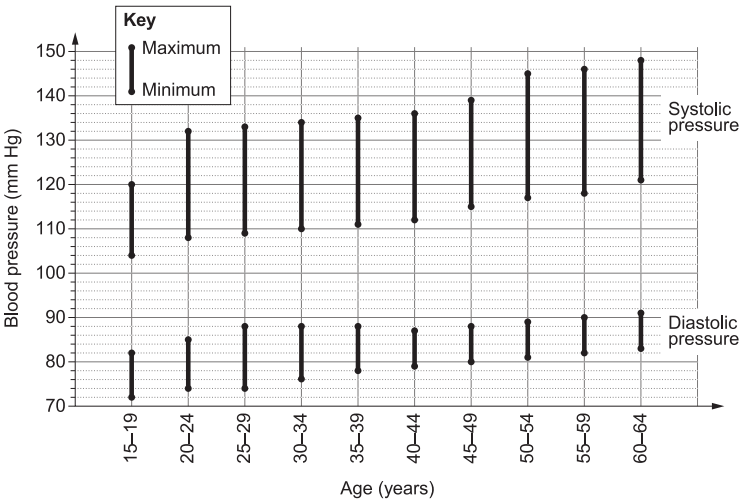
A patient is to receive 1.8 L of pain killer medication by intravenous drip that will take 1.5 hours to administer.

Given 1 mL = 4 drops, calculate the amount of drops per minute the machine must be set on. (2 marks)

20. Measurement, STD2 M7 SM-Bank 4

Blood pressure is measured using two numbers: systolic pressure and diastolic pressure. If the measurement shows 120 systolic and 80 diastolic, it is written as 120/80.

The bars on the graph show the normal range of blood pressure for people of various ages.



- a. What is the normal range of blood pressure for a 53-year-old? (2 marks)
- b. Ralph, aged 53, had a blood pressure reading of 173 over 120. A doctor prescribed Ralph a medication to reduce his blood pressure to be within the normal range. To check that the medication was being effective, the doctor measured Ralph's blood pressure for 10 weeks and recorded the following results.

Week	1	2	3	4	5	6	7	8	9	10
Blood pressure	175/120	175/120	165/110	149/98	145/92	140/90	138/88	136/88	130/85	126/84

With reference to the data provided, comment on the effectiveness of the medication during the 10-week period in returning Ralph's blood pressure to the normal range. (3 marks)

21. Measurement, STD2 M7 2017 HSC 26b

Toby's mobile phone plan costs \$20 per month, plus the cost of all calls. Calls are charged at the rate of 70 cents per 30 seconds, or part thereof. There is also a call connection fee of 50c per call.

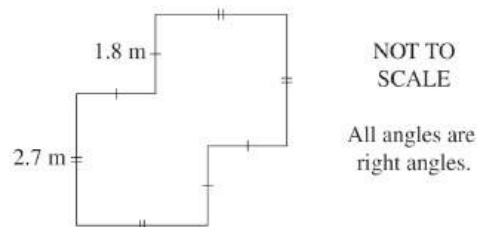
Here is a record of all his calls in July.

Date	Call duration
5 July	20 seconds
12 July	40 seconds
23 July	2 minutes 15 seconds

How much is Toby's mobile phone bill for July? (2 marks)

22. Measurement, STD2 M1 2009 HSC 23c

The diagram shows the shape and dimensions of a terrace which is to be tiled.



- Find the area of the terrace. (2 marks)
- Tiles are sold in boxes. Each box holds one square metre of tiles and costs \$55. When buying the tiles, 10% more tiles are needed, due to cutting and wastage.
Find the total cost of the boxes of tiles required for the terrace. (2 marks)

23. Measurement, STD2 M7 2008 HSC 23c

An alcoholic drink has 5.5% alcohol by volume. The label on a 375 mL bottle says it contains 1.6 standard drinks.

- How many millilitres of alcohol are in a 375 mL bottle? (1 mark)
- It is recommended that a fully-licensed male driver should have a maximum of one standard drink every hour.
Express this as a rate in millilitres per minute, correct to one decimal place. (2 marks)

24. Measurement, STD2 M7 2018 HSC 28c

Every day, a 1200-watt microwave oven is used for 45 minutes at 40% power. Electricity is charged at \$0.25 per kWh.

What is the cost of running this microwave oven for 180 days? (3 marks)

25. Measurement, STD2 M7 SM-Bank 1

Bikram runs a hot yoga studio.

If it costs 34 cents for 1-kilowatt (1000 watts) for 1 hour, how much does it cost him to run three 3200-watt heaters from 9:00 am to 12:30 pm on a single day? (Give your answer to the nearest cent) (2 marks)

26. Measurement, STD2 M7 SM-Bank 16

A refreshment stand at the market sells its iced tea in four different sizes.

			
150 mL for \$1.50	750 mL for \$6.10	375 mL for \$3.00	200 mL for \$1.80

Showing calculations, find which serving size offers the **cheapest** iced tea. (2 marks)

[+ Show Answers Only](#)

27. Measurement, STD1 M4 2020 HSC 12

Two painters each provide a quote for painting an area of 1500 square metres. Painter A charges \$100 per 30 square metres. Painter B charges \$80 per hour and bases their quote on painting 25 square metres per hour.

Calculate how much will be saved by choosing the cheaper quote. (3 marks)

28. Measurement, STD1 M4 2021 HSC 11

Chocolate of a particular brand can be bought in three different sizes.

Option 1: 100 grams for \$1.50

Option 2: 300 grams for \$4.20

Option 3: 500 grams for \$7.25

Which option gives the lowest price per 100 grams? Justify your answer with calculations. (2 marks)

29. Measurement, STD2 M7 SM-Bank 14

Reece looks at the price of four sunscreens.



Sun sunscreen
\$10.90, 1200 mL



Sport sunscreen
\$5.40, 600 mL



Block sunscreen
\$4.75, 500 mL

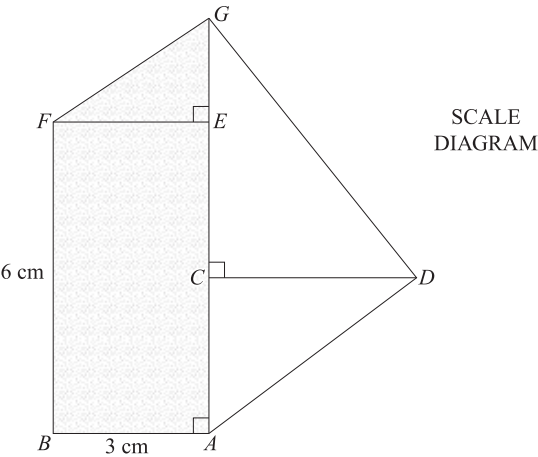


Wet sunscreen
\$7.40, 750 mL

Which sunscreen is the cheapest per litre? Show your calculations. (2 marks)

30. Measurement, STD2 M7 2018 HSC 26g

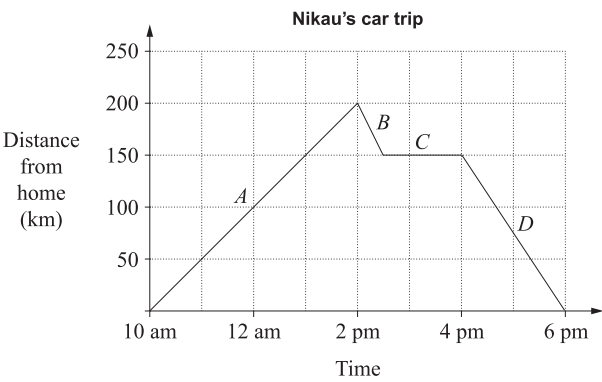
A field diagram of a block of land has been drawn to scale. The shaded region **ABFG** is covered in grass.



- The actual length of **AG** is 24 m.
- i. If the length of **AG** on the field diagram is 8 cm, what is the scale of the diagram? (1 mark)
 - ii. How much fertiliser would be needed to fertilise the grassed area **ABFG** at the rate of 26.5 g /m²? (3 marks)

31. Measurement, STD1 M4 2019 HSC 18

The travel graph displays Nikau's car trip along a straight road from home and back again. The trip has been broken into four separate sections: **A**, **B**, **C** and **D**.



- a. How far did Nikau travel in total? (1 mark)
- b. In which section of the trip, **A**, **B**, **C** and **D**, did Nikau travel the fastest? (1 mark)

32. Measurement, STD2 M7 2021 HSC 27

The price and the power consumption of two different brands of television are shown.

Television A	Television B
Price: \$900	Price: \$921.90
Power: 176 W	Power: 160 W

- The average cost for electricity is 25c/kWh. A particular family watches an average of 3 hours of television per day.
- a. The annual cost of electricity for Television A for this family is \$48.18.
For this family, what is the difference in the annual cost of electricity between Television A and Television B? (2 marks)
 - b. For this family, how many years will it take for the total cost of buying and using Television A to be equal to the cost of buying and using Television B? (2 marks)

33. Measurement, STD2 M1 2013 HSC 26d

A section of Jim's electricity bill is shown.

Energy Used and Costs					
METER ID	THIS READING	LAST READING	ENERGY USED	RATE (per kWh)	COST
Peak Energy Charge					
TMV04221/01	531.2	274.8	256.4 kWh	47.7700c	\$122.48
Shoulder Energy Charge					
TMV04221/02	A	560.9	523.5 kWh	19.4000c	\$101.56
Off-peak (Night) Energy Charge					
TMV04221/03	242.5	0.0	242.5 kWh	9.6000c	\$23.28

- What is the value of **A**? (1 mark)
- How much will Jim save if he uses 154 kWh of energy at the Off-peak rate rather than at the Peak rate? (2 marks)

34. Measurement, STD2 M1 2015 HSC 30a

The energy consumption of a computer in standby mode is 21 watts. The cost of electricity is 31 cents per kWh.

A school computer room has 20 computers.

How much will the school save by switching off all 20 computers during 11 weeks of school holidays? (2 marks)

35. Measurement, STD1 M4 2021 HSC 22

A tap is leaking water. It leaks 1 drop every 4 seconds, and 15 of these drops make up 1 mL.

- Find the amount of water leaked in a 24-hour period. Give the answer in litres. (3 marks)
- A bucket can hold 9 litres of water. How long will it take for the leaking tap to completely fill this empty bucket? (1 mark)

Worked Solutions

1. Algebra, STD2 A2 2017 HSC 14 MC

Fuel used by car **A**

$$= \frac{8000}{100} \times 9$$

$$= 720 \text{ L}$$

Fuel used by car **B**

$$= \frac{8000}{100} \times 3.5$$

$$= 280 \text{ L}$$

$\therefore$ Fuel saved using car **B**

$$= 720 - 280$$

$$= 440 \text{ L}$$

$\Rightarrow$ **B**

2. Measurement, STD2 M7 2016 HSC 11 MC

Volume required

$$= \frac{2000}{100} \times 5$$

$$= 100 \text{ mL}$$

$\Rightarrow$ **C**

3. Measurement, STD2 M7 2016 HSC 9 MC

Water used by old machine

$$= 130 \times 2 \times 52$$

$$= 13\,520 \text{ L}$$

Water used by new machine

$$= 50 \times 2 \times 52$$

$$= 5200 \text{ L}$$

$\therefore$ Water saved = 13 520 – 5200

$$= 8320 \text{ L}$$

$\Rightarrow$ **D**

4. Algebra, STD2 A2 2018 HSC 5 MC

Total distance travelled each week

$$= 5 \times 2 \times 20$$

$$= 200 \text{ km}$$

$\therefore$ **Total fuel used**

$$= \frac{200}{100} \times 8 \text{ L}$$

$$= 16 \text{ L}$$

$\Rightarrow$ **A**

5. Measurement, STD1 M4 2019 HSC 6 MC

Annie has both measures in the healthy range.

$\Rightarrow$ **D**

♦ Mean mark 48%.

6. Measurement, STD2 M4 2019 HSC 3 MC

Price per kilogram:

$$100 \text{ g} \rightarrow 10 \times 0.40 = \$4.00$$

$$500 \text{ g} \rightarrow 2 \times 1.65 = \$3.30$$

$$1 \text{ kg} \rightarrow \$3.50$$

$$2 \text{ kg} \rightarrow 6.90 \div 2 = \$3.45$$

$\Rightarrow$ **B**

7. Algebra, STD2 A2 SM-Bank 4 MC

40 litres are used to travel $3.5 \times 100 \text{ km}$

Fuel consumption (L/100 km)

$$= \frac{40}{3.5}$$

$$= 11.428$$

$$= 11.4 \text{ L/100 km}$$

$\Rightarrow$ **D**

8. Algebra, STD2 A2 2014 HSC 22 MC

Fuel used in short distance

$$= \frac{65}{100} \times 8.9 \text{ L} = 5.785 \text{ L}$$

Fuel used in long distance

$$= \frac{495}{100} \times 6.6 \text{ L} = 32.67 \text{ L}$$

$\therefore$ **Total Fuel = $5.785 + 32.67$**

$$= 38.455 \text{ L}$$

$\Rightarrow$ **B**

9. Measurement, STD2 M7 2014 HSC 17 MC

Paracetamol needed

$$= \frac{14}{2} \times 15 \text{ mg}$$

$$= 105 \text{ mg}$$

Since 120 mg is contained in 10 mL,

$\Rightarrow$ **105 mg is contained in**

$$\frac{105}{120} \times 10 \text{ mL} = 8.75 \text{ mL}$$

$\Rightarrow$ **B**

10. Measurement, STD2 M7 SM-Bank 15 MC

Convert to price per kg:

Packet 1

$$200 \text{ g} = \$1.80$$

$$1 \text{ kg} = 5 \times 1.80 = \$9$$

Packet 2

$$750 \text{ g} = \$9$$

$$100 \text{ g} = 9.00 \div 7.5 = \$1.20$$

$$1 \text{ kg} = 1.20 \times 10 = \$12$$

$\therefore$ Difference per kg

$$= 12 - 9$$

$$= \$3.00$$

$\Rightarrow B$

11. Measurement, STD2 M7 2016 HSC 15 MC

5×30 second blocks

$\therefore$ Cost = 5×0.54

$$= \$2.70$$

$\Rightarrow D$

♦ Mean mark 44%.

12. Measurement, STD2 M1 2014 HSC 20 MC

$$\text{Usage per day} = 4 \times 13 = 52 \text{ mins}$$

$$\text{Usage per week} = 7 \times 52 = 364 \text{ mins}$$

Converting to kWh

$$= (\text{hours of usage}) \times 9 \text{ kW}$$

$$= \frac{364}{60} \times 9$$

$$= 54.6 \text{ kWh}$$

$\therefore$ Cost = $54.6 \times 11.97\text{c}$

$$\approx 654\text{c}$$

$$\approx \$6.54$$

$\Rightarrow B$

♦ Mean mark 39%.

13. Measurement, STD1 M4 2019 HSC 5 MC

$$3 \text{ m/min} = 3 \times 100 \text{ cm/minute}$$

$$= 3 \times 100 \div 60 \text{ cm/second}$$

$\Rightarrow A$

♦ Mean mark 47%.

14. Measurement, STD2 M7 2018 HSC 27a

$$\text{SMS charge} = 120 \times 33\text{c} = \$39.60$$

$$\text{Data charge} = 1400 \times 26\text{c} = \$364.00$$

$\therefore$ Total bill = $50 + 39.60 + 364$

$$= \$453.60$$

15. Algebra, STD2 A2 2016 HSC 26c

Fuel used in country

$$= 170 \times \frac{5.9}{100}$$

$$= 10.03 \text{ L}$$

Fuel used in city

$$= 25 \times \frac{7.3}{100}$$

$$= 1.825 \text{ L}$$

$\therefore$ Total fuel used

$$= 10.03 + 1.825$$

$$= 11.855 \text{ L}$$

16. Algebra, STD2 A2 2014 HSC 27b

Compare cost of a 1-way trip

Motorcycle

$$\text{Fuel used} = \frac{34}{17} = 2 \text{ L}$$

$$\text{Cost} = 2 \times \$1.67 = \$3.34$$

Bus

$$\text{Cost} = \frac{36.40}{10} = \$3.64$$

$$\text{Difference} = \$3.64 - 3.34 = \$0.30$$

$\therefore$ Motorcycle is \$0.30 cheaper per 1-way trip.

17. Measurement, STD2 M7 2015 HSC 27b

Fluid rate of delivery

$$= \frac{2400}{12}$$

$$= 200 \text{ mL per hour}$$

$$= \frac{200}{60}$$

$$= 3\frac{1}{3} \text{ mL per minute}$$

Since each mL has 15 drops

$$\# \text{ Drops} = 15 \times 3\frac{1}{3}$$

$$= 50 \text{ per minute}$$

18. Measurement, STD2 M1 2016 HSC 28b

Cost to run heater for 5 years

$$= 5 \times 505 \times 0.35$$

$$= \$883.75$$

$\therefore$ Total purchase and running cost

$$= 883.75 + 990$$

$$= \$1873.75$$

19. Measurement, STD2 M7 SM-Bank 8

$$\begin{aligned} \text{Total drops required} &= 1800 \times 4 \\ &= 7200 \text{ drops} \end{aligned}$$

$$\begin{aligned} \text{Time (in minutes)} &= 1.5 \times 60 \\ &= 90 \text{ minutes} \end{aligned}$$

$$\begin{aligned} \text{Drops per minute} &= \frac{7200}{90} \\ &= 80 \end{aligned}$$

$\therefore$ The machine must be set to 80 drops per minute.

20. Measurement, STD2 M7 SM-Bank 4

a. Normal range (systolic): 117 - 145

Normal range (diastolic): 81 - 89

b. Effectiveness of medication

- systolic and diastolic pressure reduced below original readings from week 3 onwards.
- systolic pressure in normal range from week 5 onwards.
- diastolic pressure in normal range from week 7 onwards.
- blood pressure only in the normal range from week 7.

21. Measurement, STD2 M7 2017 HSC 26b

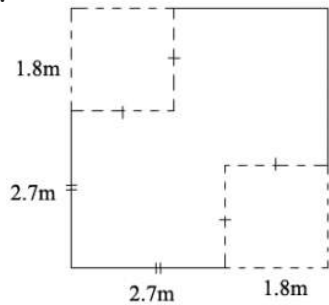
$$\text{Call cost} = 0.70 + (2 \times 0.70) + (5 \times 0.70) = \$5.60$$

$$\text{Connection fees} = 3 \times 0.50 = \$1.50$$

$$\begin{aligned} \therefore \text{Total bill} &= 5.60 + 1.50 + 20 \\ &= \$27.10 \end{aligned}$$

22. Measurement, STD2 M1 2009 HSC 23c

i.



$$\begin{aligned}
 \text{Area} &= \text{Area of big square} - \text{Area of 2 cut-out squares} \\
 &= (2.7 + 1.8) \times (2.7 + 1.8) - 2 \times (1.8 \times 1.8) \\
 &= 20.25 - 6.48 \\
 &= 13.77 \text{ m}^2
 \end{aligned}$$

ii. Tiles required = $(13.77 + 10\%) \times 13.77$
 $= 15.147 \text{ m}^2$

$\Rightarrow$ 16 boxes are needed

$$\begin{aligned}
 \therefore \text{Total cost of boxes} &= 16 \times \$55 \\
 &= \$880
 \end{aligned}$$

23. Measurement, STD2 M7 2008 HSC 23c

i. Alcohol = $\frac{5.5}{100} \times 375$
 $= 20.625 \text{ mL}$

ii. Since 1.6 standard drinks = 375 mL
 $\Rightarrow$ 1 standard drink
 $= \frac{375}{1.6}$
 $= 234.375 \text{ mL}$

$$\begin{aligned}
 \therefore \text{Rate} &= \frac{234.375}{60} \\
 &= 3.90625 \\
 &= 3.9 \text{ mL/min}
 \end{aligned}$$

24. Measurement, STD2 M7 2018 HSC 28c

$$\begin{aligned}
 \text{Daily usage} &= 1200 \times \frac{45}{60} \times 40\% \\
 &= 360 \text{ watts}
 \end{aligned}$$

$$\begin{aligned}
 180 \text{ day usage} &= 180 \times 360 \\
 &= 64\,800 \text{ watts} \\
 &= 64.8 \text{ kW}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Cost over 180 days} &= 64.8 \times 0.25 \\
 &= \$16.20
 \end{aligned}$$

25. Measurement, STD2 M7 SM-Bank 1

Total energy usage

$$= 3 \times 3200 \times 3.5 \text{ hours}$$

$$= 33\,600 \text{ watts}$$

$$\therefore \text{Cost} = \frac{33\,600}{1000} \times 0.34$$

$$= 11.424$$

$$= \$11.42 \text{ (nearest cent)}$$

26. Measurement, STD2 M7 SM-Bank 16

Consider the price per mL of each option

$$150 \text{ mL} : \frac{1.50}{150} = 1 \text{ cent/mL}$$

$$750 \text{ mL} : \frac{6.10}{750} = 0.81 \text{ cent/mL}$$

$$375 \text{ mL} : \frac{3.00}{375} = 0.8 \text{ cent/mL}$$

$$200 \text{ mL} : \frac{1.80}{200} = 0.9 \text{ cent/mL}$$

$\therefore$ 375 mL serving size offers the cheapest price

27. Measurement, STD1 M4 2020 HSC 12

Painter A cost:

$$C_A = \frac{1500}{30} \times 100 = \$5000$$

Painter B cost:

$$\text{Hours} = \frac{1500}{25} = 60$$

$$C_B = 60 \times 80 = \$4800$$

$$\therefore \text{Savings by using Painter B}$$

$$= 5000 - 4800$$

$$= \$200$$

28. Measurement, STD1 M4 2021 HSC 11

Compare prices per 100 grams:

$$\text{Option 1} = \$1.50$$

$$\text{Option 2} = 4.20 \div 3 = \$1.40$$

$$\text{Option 3} = 7.25 \div 5 = \$1.45$$

$\therefore$ Option 2 is the cheapest.

29. Measurement, STD2 M7 SM-Bank 14

$$\text{Sun} = \frac{1090}{1200} = 0.91 \text{ c/mL}$$

$$\text{Sport} = \frac{540}{600} = 0.9 \text{ c/mL}$$

$$\text{Block} = \frac{475}{500} = 0.95 \text{ c/mL}$$

$$\text{Wet} = \frac{740}{750} = 0.99 \text{ c/mL}$$

$\therefore$ Sport sunscreen is the cheapest.

STRATEGY: Finding the cost per mL is much more efficient in answering this question than finding the cost per litre.

30. Measurement, STD2 M7 2018 HSC 26g

i. Scale 8 cm : 24 m

$$1 \text{ cm} : 3 \text{ m}$$

$$1 : 300$$

♦♦ Mean mark 32%.

ii. Area of rectangle $ABFE$

$$= 6 \text{ cm} \times 3 \text{ cm}$$

$$= 18 \text{ m} \times 9 \text{ m}$$

$$= 162 \text{ m}^2$$

Area of $\triangle EFG$

$$= \frac{1}{2} \times 3 \text{ cm} \times 2 \text{ cm}$$

$$= \frac{1}{2} \times 9 \times 6$$

$$= 27 \text{ m}^2$$

$$\therefore \text{Fertiliser needed} = (162 + 27) \times 26.5 \\ = 5008.5 \text{ grams}$$

31. Measurement, STD1 M4 2019 HSC 18

a. Distance travelled = 2×200

$$= 400 \text{ km}$$

b.

♦ Mean mark part (b) 42%.

Fastest section has the steepest slope (in either direction).

$\therefore$ Section D was the fastest.

32. Measurement, STD2 M7 2021 HSC 27

a. Annual power usage (B) = $160 \times 3 \times 365$

$$= 175\,200$$

$$= 175.2 \text{ kWh}$$

Annual cost (B) = 175.2×0.25

$$= \$43.80$$

♦ Mean mark part (a) 49%.

Difference in cost = $48.18 - 43.80$

$$= \$4.38$$

b. Difference in price = $921.90 - 900$

$$= \$21.90$$

♦♦ Mean mark part (b) 23%.

$$\text{Years to even out cost} = \frac{21.90}{4.38}$$

$$= 5 \text{ years}$$

33. Measurement, STD2 M1 2013 HSC 26d

i. $A = \text{Last reading} + \text{Energy used}$

$$= 560.9 + 523.5$$

$$= 1084.4$$

♦ Mean mark part (i) 43% and part (ii) 46%.

ii. Cost at off-peak = 154×9.6

$$= 1478.4 \text{ cents}$$

Cost at peak = 154×47.77

$$= 7356.58 \text{ cents}$$

$\therefore$ Saving = $7356.58 - 1478.4$

$$= 5878.18$$

$$= \$58.78 \text{ (nearest cent)}$$

34. Measurement, STD2 M1 2015 HSC 30a

21 watts usage per computer per hour.

Watts used by 20 computers in 11 weeks

$$= 21 \times 20 \times 24 \times 7 \times 11$$

$$= 776\,160 \text{ watts}$$

$$= 776.16 \text{ kW}$$

$\therefore$ Cost of energy

$$= 776.16 \times \$0.31$$

$$= \$240.6096$$

$$= \$240.61 \text{ (nearest cent)}$$

$\therefore$ The school will save \$240.61 by switching off all the computers.

◆ Mean mark 49%.

35. Measurement, STD1 M4 2021 HSC 22

a. Drops per minute = $\frac{60}{4} = 15$

Volume per minute = 1 mL

$\therefore$ Volume in 24 hours = $1 \times 60 \times 24$

$$= 1440 \text{ mL}$$

$$= 1.44 \text{ L}$$

◆ Mean mark part (a) 50%.

b. Time to fill bucket = $\frac{9}{1.44}$

$$= 6.25 \text{ days}$$

◆◆ Mean mark part (b) 32%.